Evidence report: Smartphones, Parents, and Adolescent Wellbeing in Colombia

Pilot findings in support of the main RCT

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1. Summary

- **The problem.** Adolescent smartphone use, and time on social-media apps in particular, has risen sharply worldwide. The closest causal evidence links social-media access to worse young adult mental health (Braghieri, Levy, and Makarin 2022, *AER*). Governments have responded primarily through school-based restrictions, but most adolescent phone use happens at home, where parents in low- and middle-income settings have few scripts or tools to guide it.
- **The evidence for a parent-focused design.** In our 2025 baseline (350 adolescents, four Bogotá public schools), students average **5.4 hours** of phone use on a typical day and **8.4 hours** on the heaviest day of the past week. Heavier use correlates with worse mental-health symptoms ($r = 0.31$), less sleep, and lower grades. Students whose parents set at least one household phone rule use phones **1.1 fewer hours per day** ($d = 0.34$, $p = 0.002$); parental rule-setting is a stronger correlate of lower use than any student-side variable we measured. A prior student-directed classroom pilot produced no detectable reduction in phone use on any main outcome, though students did learn the content, consistent with the conclusion that the binding lever sits in the home rather than the classroom.
- **The program.** *Mi Ritmo Digital* is a seven-call parental coaching sequence built around two gaps we document in the pilot: parents underestimate their child's phone use (a belief gap), and want to set household rules but lack scripts for doing so (a capability gap). The program closes both gaps by pairing structured coaching on sleep, screen time, and content quality with a live moment in which the parent reads their child's actual app usage off our passive-tracking app. Since February 2026 we have been delivering the program to 100 Bogotá families. Qualitative review of the first group shows no dropouts after Call 1, active engagement across calls, and specific, dated commitments at the close of each call (bedtime phone-collection times, per-app limits, airplane-mode routines). These observations are descriptive; the main RCT is designed to produce causal estimates.
- **The main RCT.** A 1,200-family RCT in Bogotá public schools will test the program against a waitlist control. Four pre-registered co-primary outcome families (with multiple-testing correction) cover phone use (measured passively via our secure Android app), mental health,

learning outcomes, and parental practices.

- **Scale-up.** The Secretaría de Educación de Bogotá has expressed strong interest in the program, including an offer of funding toward delivery, as it prepares to implement a pending phones-in-schools law. Scale-up beyond Bogotá, conditional on RCT results, would proceed through established government channels, drawing on co-author Juan Saavedra's (Harvard) experience delivering education programs at national scale in Colombia.

2. Motivation and Colombian context

Smartphone adoption among adolescents has outpaced the parenting norms, school policies, and evidence base that would usually accompany a new technology. The closest causal evidence on harm comes from Braghieri, Levy, and Makarin (2022, *AER*), who exploit the staggered rollout of Facebook across US colleges and estimate that access raised clinically relevant depression by 9% and severe anxiety by 12%, with effects concentrated among women. LMIC evidence is thinner but directionally consistent: Jain and Stemper (2024) find that 3G rollout reduced test scores by roughly 0.06 SD across low- and middle-income countries. On the supply-side question of what parents can do, established parent-training programmes on adjacent outcomes (Triple P, Sanders et al. 2014; Familias Unidas) typically generate effect sizes of d = 0.4–0.6, consistent with broader meta-analytic evidence that brief, structured parent-training produces robust behavioural change across settings (Michelson et al., 2013). These two literatures together motivate this study: if phone use causally harms adolescent wellbeing, and if parental behaviour is malleable through short, structured coaching, then a parent-focused program has a plausible path to impact.

Colombia is a policy-relevant setting to run this study. Smartphone penetration exceeds 80% among grade 7–10 students in Bogotá public schools, where our 2025 baseline shows average daily use of 5–6 hours (Section 3). Girls spend roughly 1.5 hours more per day on their phones than boys and score higher on standard depression indices. The Colombian Congress is currently debating a national phones-in-schools law, and the Secretaría de Educación de Bogotá (SED) is seeking rigorous evidence on family-side interventions before the law is finalised. Bogotá's 430 public schools serve over 800,000 students; grades 7–10 alone cover roughly 150,000 families, which defines a concrete near-term scale-up target.

We focus on parents for two reasons. First, most adolescent phone use occurs outside school. Governments have responded primarily through school-based measures: Australia and France have banned social media for under-16s, and several US states and Colombia are moving toward phones-in-schools bans. These measures address only a small share of daily exposure, and arguably a shrinking share as children displace use from school into the home. Second, the two specific gaps that the program is designed to close both sit at the household level. Parents in our pilot calls routinely describe their child's actual phone-use numbers as surprising, which we label a *belief gap*: parents cannot set informed rules if they do not know what their child is actually doing. Parents also report wanting household rules but lacking concrete ideas, scripts, data, and follow-through to set them, which we label a *capability gap*. The coaching sequence closes both gaps: it provides accurate information from a passive-tracking app, and practical rule-setting skills from a short, structured call sequence.

3. Baseline evidence from the 2025 endline: adolescents use their phones a lot

This section establishes the two factual preconditions for the intervention: that adolescent phone use in our setting is heavy, and that it moves with the wellbeing outcomes the program targets. In November–December 2025 we surveyed 350 adolescents across four Bogotá public schools (grades 8–10). Smartphone use in this sample is heavy, skewed toward short-form video, and correlated with mental-health symptoms, sleep loss, and lower school grades.

On a typical day, students report using their phone for **5.4 hours** (median 5.0, SD 3.6, n = 350). On their heaviest day in the past week, the average rises to **8.4 hours** (Figure 1, panel A).[1] Heavy use is not a tail phenomenon: more than half the sample report over five hours of typical daily use.

Girls report higher use than boys: 6.1 versus 4.5 hours on a typical day (a 1.6-hour gap), widening to 9.1 versus 7.4 on peak days. The platform mix differs by gender (panel B): TikTok is the most-used app for **48% of girls** and 35% of boys; gaming is the most-used app for **25% of boys** and 3% of girls.

Phone use is associated with higher mental-health symptom scores. We measure symptoms with a standardised index of four items (restlessness, sadness, body-image concerns, and difficulty concentrating) asked about the past week. Across the full baseline sample, hours of phone use and the symptom index correlate at $\mathbf{r = 0.31}$ ($p < 10^{-8}$, n = 350); each additional hour of daily use is associated with **0.086 SD** worse symptoms (panel C).[2] On the endline version of the same index, girls score substantially higher than boys (1.12 SD vs. 0.58 SD; $p < 0.001$); we pre-specify gender as a heterogeneity dimension in the main RCT.

Sleep moves in the expected direction. Phone hours and hours slept correlate at $\mathbf{r = -0.16}$ ($p = 0.003$, panel D); average sleep is 7.2 hours per night and **54% of students sleep fewer than 8 hours**. These cross-sectional patterns do not identify a causal channel, but they are consistent with the accounts parents have given during our 2026 pilot calls, in which phone use at bedtime is described as a source of sleep loss.

Academic performance is also associated with phone use. In a subset of 136 students for whom we have linked school-grade records, each additional hour of daily phone use is associated with a **0.013-point decrease in average grade** (SE = 0.006, $p < 0.05$) after demographic and school-grade fixed-effects controls.[3]

These patterns are descriptive, not causal. Taken together, they establish that the problem is large enough to matter in this setting, and that the outcomes the program targets do move with phone use. What Section 3 does not answer is which household-level variable is likely to shift that use. Section 4 takes up that question.

[1] Peak-day distribution in Panel A is computed on the subsample that completed the relevant module (n = 203); typical-day, MH, and sleep analyses use the full n = 350.

[2] Baseline descriptives computed from the cleaned endline CSVs; MH index regression from the baseline phone-use/mental-health cross-section; endline gender means from the cleaned gender-by-module file.

[3] Grade-phone-use regression from `baseline_grade_avg_c1.tex`; outcome is the student's average school grade, specification uses demographic controls and school-grade fixed effects, n = 136 with linked records.

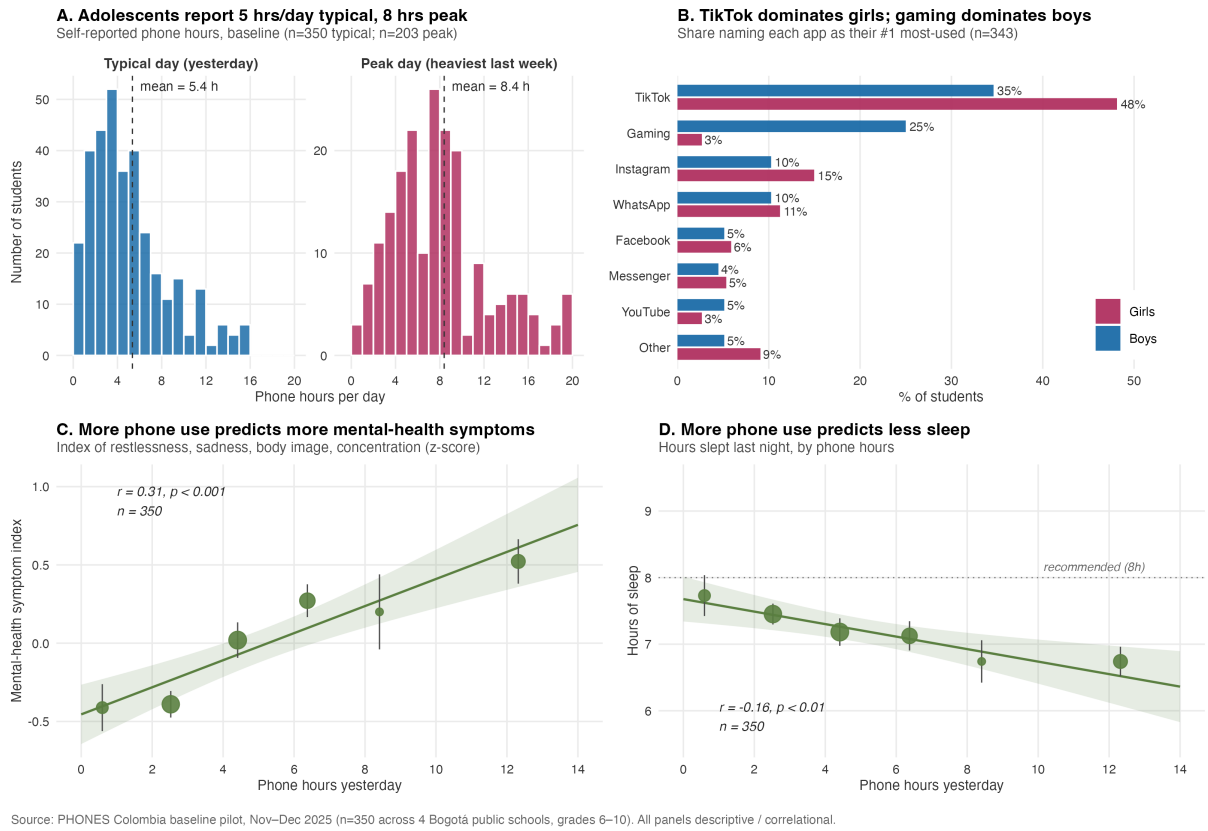

Figure 1: Figure 1. Baseline phone use among 350 Colombian public-school adolescents, Nov–Dec 2025. (A) Self-reported hours on a typical day (n = 350, mean 5.4) and heaviest day in the past week (n = 203, mean 8.4). (B) Most-used app by gender. (C) Daily phone hours vs. standardized MH symptom index (r = 0.31). (D) Daily phone hours vs. hours slept (r = −0.16). All panels are descriptive/correlational.

4. Parental rules predict sharply lower phone use

Parental rules predict substantially lower adolescent phone use, more strongly than any student-side variable we measured. Students in households with at least one phone rule use phones **1.1 fewer hours on a typical day** (4.81 vs. 5.90 raw means; d = 0.34, p = 0.002) and **1.8 fewer hours on their heaviest day** (7.18 vs. 8.97; d = 0.43, p < 0.001).[4] Fifty-seven percent of households in the endline sample (n = 335) set at least one rule. The peak-day gap is robust to demographic controls, school-grade fixed effects, and treatment-status controls (coefficients stable between −1.6 and −1.9 hrs/day; p < 0.01); the typical-day gap attenuates to −0.7 to −1.0 hrs/day (p = 0.05–0.06) under the same controls. These are cross-sectional associations, not treatment effects, and selection into rule-setting is likely non-random.

Time-based rules are associated with the largest gaps (Figure 2, panel B; each coefficient is from a

[4]Panel A raw-mean gaps on the endline sample (n = 335). With demographic controls + school-grade FE + treatment-status control, peak-day coefficients remain at −1.6 to −1.9 hrs/day (p < 0.01); typical-day coefficients attenuate to about −0.7 to −1.0 (p = 0.05–0.06). Individual-rule coefficients in Panel B are each from their own regression with demographic controls clustered at classroom; per-rule working n ranges 260–290 depending on item missingness. Phone-ownership heterogeneity is the owns-phone main effect from the peak-day heterogeneity specification; the rules × ownership interaction is not estimated in this table.

separate regression with demographic controls). "Time limit on certain apps" is associated with **−1.60 hrs/day** ($p = 0.002$), "time limit per day" with **−1.23** ($p < 0.001$), and "no phones at school" with **−1.10** ($p = 0.001$). Rules that structure when or how apps are used, rather than punitive rules such as phone confiscation, carry the larger associations. The coaching curriculum accordingly focuses on these rule types and equips parents with scripts for setting and maintaining them. These correlations were what first motivated an intervention on this margin; in 2025, we tested a classroom-based version delivered to students. Section 5 reports what happened.

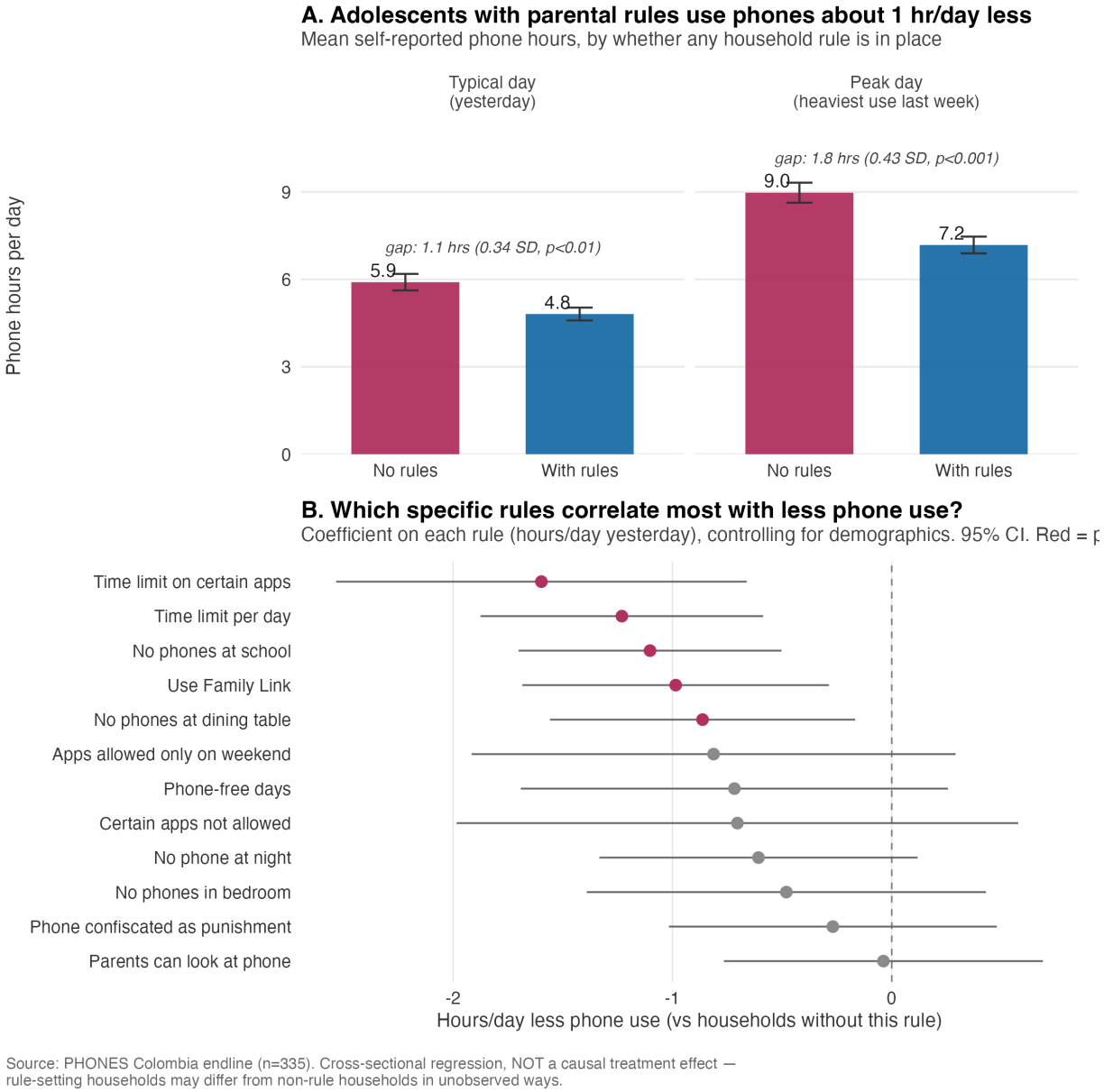

Figure 2: Figure 2. Parental rules and adolescent phone use, 2025 endline ($n = 335$). (A) Raw gaps in self-reported hours between households with at least one rule and households with none (correlational). (B) Coefficient from a separate regression per rule with demographic controls; time-based rules dominate. *Correlational — NOT a causal treatment effect.*

5. The 2025 classroom pilot: no detectable effect on phone use

Our 2025 pilot tested a student-directed version of the program, delivered as a short classroom curriculum. Treatment effects on every main phone-use outcome were small and statistically indistinguishable from zero. We read this null as informative rather than purely disappointing: it locates the binding constraint outside the classroom, and it is the reason we pivoted in 2026 to a parent-facing design.

Two reasons are consistent with the null. First, asking adolescents alone to resist deliberately engineered, habit-forming technologies is a demanding ask. The implementing partner's qualitative report flags "the brevity of the intervention, limited follow-up capacity, and lack of family support" as the recurring obstacles to students carrying classroom content into daily practice. Second, most adolescent phone use occurs outside school, so any classroom-bounded intervention (including the phones-in-schools ban under debate in Colombia) addresses only a small share of total exposure. A household-level intervention reaches the hours and settings where most use actually happens.

The pilot was a three-arm cluster-randomised study across four Bogotá public schools, delivered in partnership with Aulas en Paz (Princeton-affiliated). Of 864 grade 8–10 students invited, 457 enrolled (53% consent; 84% baseline completion, 74% endline retention). Treatment classrooms received three in-school sessions; an active control received alternative classroom programming; a pure control received no contact.

The table below reports the primary specification (demographic controls, imputed baseline outcome, school-grade fixed effects; N = 335):

Outcome	Control mean	Treatment coef. (SE)
Phone hours yesterday	5.5	−0.28 (0.33)
Phone hours on peak day	8.2	−0.25 (0.38)
TikTok hours yesterday	2.5	−0.20 (0.17)
WhatsApp hours yesterday	1.7	−0.02 (0.25)
Satisfaction with phone time	0.56	−0.05 (0.05)
Negative-impact index (SD)	0.00	+0.12 (0.07)

No point estimate reaches statistical significance, and the pattern is stable across endline-only, matched-sample, and alternative fixed-effects specifications. All point estimates are below 0.4 hr/day in absolute value, and 95% confidence intervals leave room for reductions of up to roughly 0.9 hr/day, so we interpret this as an absence of detectable signal rather than a precisely bounded zero. The negative-impact index coefficient is wrong-signed (+0.12 SD) but not statistically significant, and we treat it as sampling noise. The implementing partner's lead recommendation was to implement "shared digital parenting" sessions that "standardise the conversation about cell phone use at home," which is exactly the structure of the intervention evaluated in Section 6.

6. The 2026 parent-calls pilot: feasibility and suggestive mechanisms

The 2026 pilot tests whether a structured parent-coaching format will engage families in this setting and elicit the kind of concrete behavioural commitments the design requires. Since February 2026 we have delivered the seven-call program to 100 Bogotá families; all calls have been recorded with family permission, and we have conducted a qualitative review of the first group. The findings below draw on that review. They are descriptive observations whose purpose is to document feasibility and to generate hypotheses that the main RCT can test.

Retention and engagement. The program delivers seven calls per family over roughly three weeks: three content sessions and four shorter follow-ups. In the reviewed cases, no family dropped out after Call 1. The first three sessions run 35–45 minutes on average. Parents arrive at later calls with specific questions about events in the intervening days, which we take as a signal of active engagement rather than passive compliance.

Parents report learning new material. The sleep session reaches even parents who already collect their child's phone at night: specific facts about blue light, sleep latency, and notification mechanics are described as new. Content that parents flag as surprising in the first session reappears in their decision-making during the follow-up call.

Parents respond well to the program and leave with concrete commitments. Parents react positively across the reviewed cases. During the screen-time session the facilitator asks the parent to read their child's previous-week usage off our app, and parents typically express surprise at the numbers. Most parents leave the call with a specific, dated commitment: an earlier bedtime phone-collection time (e.g. 9:30 → 9:00 → 8:30 pm), airplane mode at night in place of power-off, or per-app timers configured during the call.

Parents also frequently describe the coaching as validating practices they were already attempting. Specific guidance (for example, "airplane mode, not off") paired with affirmation of existing instincts is what parents most often cite as valuable. Taken together, these qualitative observations suggest that a short, structured coaching format anchored in passive-tracking data is one that parents in this setting will engage with, and that it elicits concrete behavioural commitments of the kind the theory of change requires. Whether that engagement translates into measurable behaviour change is what the main RCT is designed to answer.

7. Objective measurement and the main RCT

The main RCT randomises 1,200 Bogotá public-school families (grades 7–10) into a treatment arm and a waitlist control (1–6 month delay, with the primary endline measured before crossover). Four co-primary outcome families are pre-registered, corresponding to the theory of change that runs from information (closing the belief gap) and coaching (closing the capability gap), through changed parental practices, to adolescent outcomes:

1. **Parental practices.** Rule presence, rule consistency, and parent-child communication about phones, from the parent survey. This is the first downstream outcome the program is designed to move.

2. **Phone use.** Objectively measured hours per day on the child's phone, using *Mi Ritmo Digital*, our passive-tracking Android app (now live in the Colombia Play Store), which logs total screen-on time and per-app usage (e.g. TikTok, Instagram, WhatsApp, YouTube, gaming). At n = 1,200 the study is powered to detect effects of ~0.15 SD, well below the 0.34 SD correlational benchmark in Section 4. Android penetration on parent phones in Bogotá public schools exceeds 85%; an iOS build is in development.
3. **Mental health.** PHQ-A, GAD-7, and SDQ. Gender is pre-specified as a heterogeneity dimension, given the baseline gap in Section 3.
4. **Learning outcomes.** In-app cognitive battery at baseline and endline, and end-of-year school grades where available.

The Secretaría de Educación de Bogotá has demonstrated strong interest in the program and offered funding toward delivery in support of its phones-in-schools strategy. Scale-up beyond Bogotá, conditional on RCT results, would proceed through established government channels, drawing on co-author Juan Saavedra's (Harvard) experience at national scale in Colombia. The design is aimed throughout at evidence that a government partner can act on: a structured, low-cost, replicable coaching format evaluated on passively measured phone use at a sample size that can detect effects that matter at scale.